

T an operator on n -dim V -space V of dim n over field F

Thm (Fitting decomposition) ① $\left. \begin{array}{l} \ker T \subset \ker T^2 \subset \dots \\ \operatorname{Im} T \supset \operatorname{Im} T^2 \supset \dots \end{array} \right\} \begin{array}{l} \text{Each chain stabilizes,} \\ \text{as soon as a term repeats.} \\ \text{no later than } T^n \text{ term} \end{array}$

$$V = \ker(T^\infty) \oplus \operatorname{Im}(T^\infty)$$

Notation $\ker(T^\infty) = \ker T^n$

$$\operatorname{Im}(T^\infty) = \operatorname{Im} T^n$$

Ex. Write full pf.
Show spanning
explicitly. Show

$\ker(T^\infty)$ is the largest T -stable
subspace on which T is nilpotent

Observe that $\ker(T^i), \operatorname{Im}(T^i)$ are T -stable $T|_{\operatorname{Im}(T^\infty)}$ is bijective

$\operatorname{Im}(T^\infty)$ is the largest T -stable
subspace on which T is an isom

Proof of $\oplus$ By rank nullity applied to T^n , we know

$$\dim(\ker T^n) + \dim(\operatorname{Im} T^n) = \dim V. \text{ So E.T.S } \ker(T^n) \cap \operatorname{Im}(T^n) = \{0\}$$

$$\text{Let } v \in \text{intersection } T^n v = 0 \text{ and } T^n w = v \Rightarrow T^n v = T^{2n} w = 0 \dots$$

$\text{Im}(T^\infty)$ is the largest T -stable subspace on which T is an isom

Proof of $\oplus$: By rank nullity applied to T^n , we know
 $\dim(\ker T^n) + \dim(\text{Im } T^n) = \dim V$. So E.T.S $\ker(T^n) \cap \text{Im}(T^n) = 0$
Let $v \in \text{intersection}$. $T^n v = 0$ and $T^n w = v \Rightarrow T^n v = T^{2n} w = 0 \dots$

Assume char poly $P_T(x) = \prod_{i=1}^k (x - \lambda_i)^{m_i}$

Note: $\sum_{i=1}^k m_i = n$ Ex: Why is the factorization unique?

Recall T is diag $\Leftrightarrow V = \bigoplus_j \ker(\lambda_j I - T)$ | Ex $\dim \ker(\lambda_j I - T) \leq m_j$

$\Leftrightarrow \dim \ker(\lambda_j I - T) = m_j \forall j$ | "geometric mult $\leq$ algebraic multiplicity"

Thm If $P_T(x) = \prod_j (x - \lambda_j)^{m_j}$, $V = \bigoplus_j \ker(\lambda_j I - T)^\infty$

can take $\infty = \dim V$

Ex: Can we take $\infty = m_j$ (for λ_j)?

Defⁿ $\ker(\lambda I - T)^\infty =$ generalized λ -eigenspace of T . Observe that gen λ -espaces of T are T -stable

Pf: You do, using. We have long division for polynomials optional
just like in $\mathbb{Z}$. As a result we have gcd of polynomials p_1, p_2 which can be obtained as gcd

For an operator T on V -space V ,
 a subspace W of V is called
 T -stable (T -invariant) if $T(W) \subset W$.
 i.e. " $T|_W$ = restriction of T to domain W "
 is an operator on W .

Let $\underbrace{w_1, \dots, w_k}_{\text{basis of } W}, \underbrace{u_1, \dots, u_\ell}_{\text{call span of these } U}$ be a basis of V
 $\therefore V = W \oplus U$.

Matrix of T wrt to this basis looks like

$$\begin{array}{c} w_1 \dots w_k \quad u_1 \dots u_\ell \\ \begin{array}{c} w_1 \\ \vdots \\ w_k \\ \hline u_1 \\ \vdots \\ u_\ell \end{array} \left[\begin{array}{cc|cc} & & & \\ & A & & B \\ & & & \\ \hline & & 0 & D \\ & & & \end{array} \right] \end{array}$$

Rmk 1-dim T -inv subspace
 $\rightsquigarrow$ an eigenvector

key question

Can I make
 $B = 0$ using
 a suitable basis?
 If that is possible
 then V is also
 T -invariant and
 we can say

$$T = T|_W \oplus T|_U$$

basis of W call span of these U
 $\therefore V = W \oplus U$

Rank 1-dim T -inv subspace
 $\leadsto$ an eigenvector

T -invariant and
 we can say
 $T = T|_W \oplus T|_U$

Jordan canonical form
 eigenspace $\ker(\lambda I - T)^\infty$

If $p_T(x) = \prod (x - \lambda_j)^{m_j}$ then each generalized
 has a basis wrt which the matrix of T
 has a block diagonal form with each of these, smaller blocks

of Jordan type $J_{\lambda, d} = \begin{bmatrix} \lambda & 1 & & 0 \\ & \lambda & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda \end{bmatrix}_{d \times d}$

Moreover the sizes of these J -blocks
 are uniquely determined by T .

$T - \lambda I$ 

$a \overline{b}$
 $b = aq + r$
 $\gcd(b, a) = \gcd(a, r)$
 $= \gcd(d, 0)$
 $|r| < |a|$